Fish Population Dynamics in Lake Constance

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29.10.2026

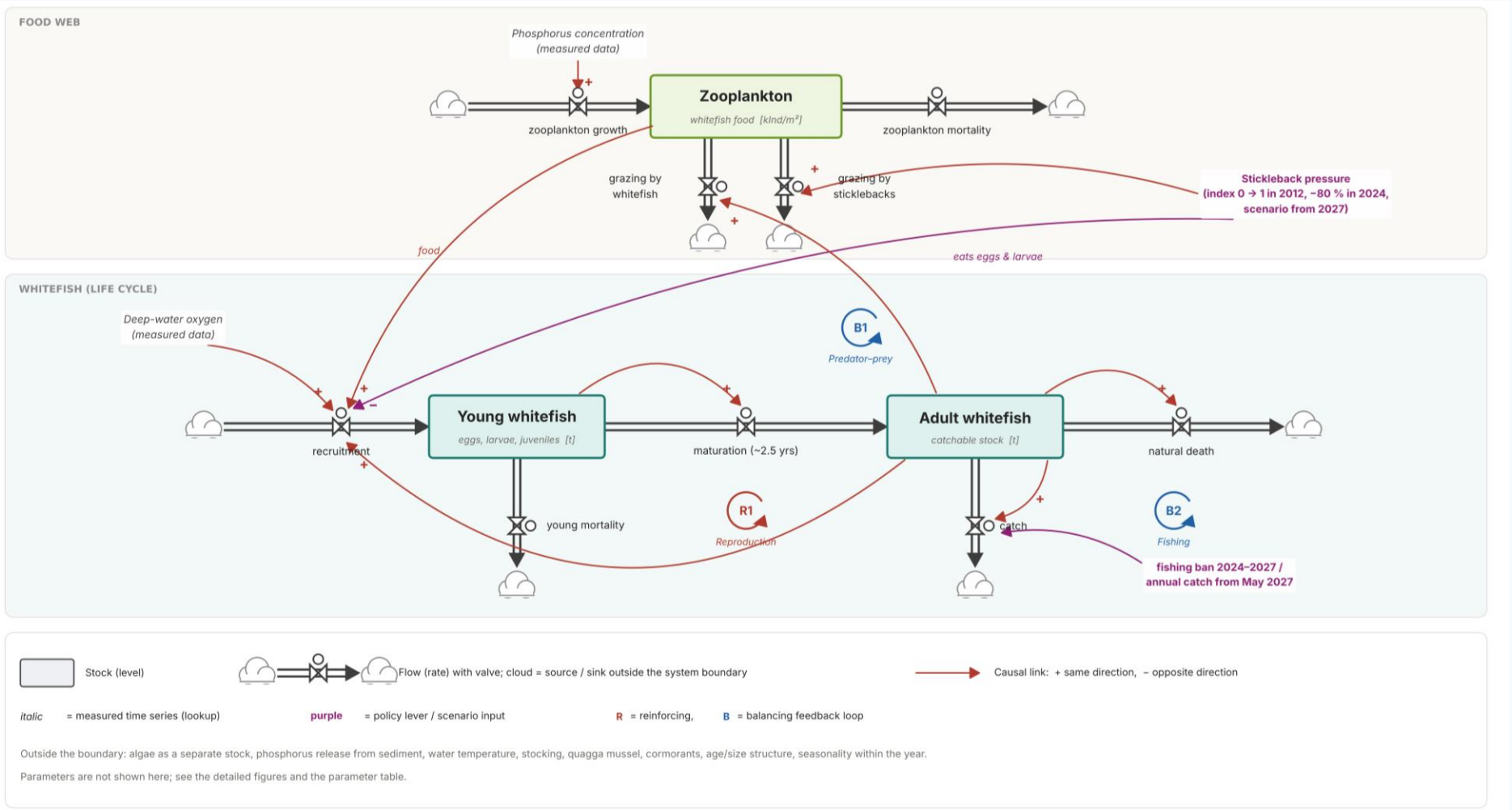
Model Idea


SRF
News
Sport
Meteo
Kultur
Wissen
Kids
Mehr Felchen im Bodensee
Kompromiss: Felchen-Fangverbot im Bodensee wird leicht gelockert
Im Bodensee hat es wieder mehr, ältere und grössere Felchen. Deshalb dürfen von Mai bis August 2027 Felchen gefischt werden. Aber es gibt strenge Auflagen.

Model Idea

- Whitefish: key role in ecosystem & important food fish
  - 1978 Peak fishing catch with 1200 tonnes
  - 2023 only 9.8 tonnes fishing
- 2024 - 2027 fishing ban
- Stickleback as rival

Model Idea


FOOD WEB
Phosphorus concentration
(measured data)
Zooplankton
whitefish food [kInd/m²]
zooplankton growth
zooplankton mortality
grazing by
whitefish
grazing by
sticklebacks
Stickleback pressure
(index 0 → 1 in 2012, −80 % in 2024,
scenario from 2027)
food
eats eggs & larvae
WHITEFISH (LIFE CYCLE)
Deep-water oxygen
(measured data)
B1
Predator–prey
Young whitefish
eggs, larvae, juveniles [t]
Adult whitefish
catchable stock [t]
recruitment
maturation (~2.5 yrs)
natural death
young mortality
R1
Reproduction
catch
B2
Fishing
fishing ban 2024–2027 /
annual catch from May 2027
Stock (level)
Flow (rate) with valve; cloud = source / sink outside the system boundary
Causal link: + same direction, − opposite direction
italic = measured time series (lookup)
purple = policy lever / scenario input
R = reinforcing, B = balancing feedback loop
Outside the boundary: algae as a separate stock, phosphorus release from sediment, water temperature, stocking, quagga mussel, cormorants, age/size structure, seasonality within the year.
Parameters are not shown here; see the detailed figures and the parameter table.

Model boundaries

- Model focuses on whitefish population in Lake Constance
- Timeframe: 1950 – 2050, seasons not included
- Water temperature/ climate change not included
- Release of phosphorus from the sediment is not included
- Stickleback only included as a pressure index, not as its own population
- Algae not modelled as a stock (included in zooplankton growth)

Hypothesis

- **H1**: The model can reproduce the historical data.

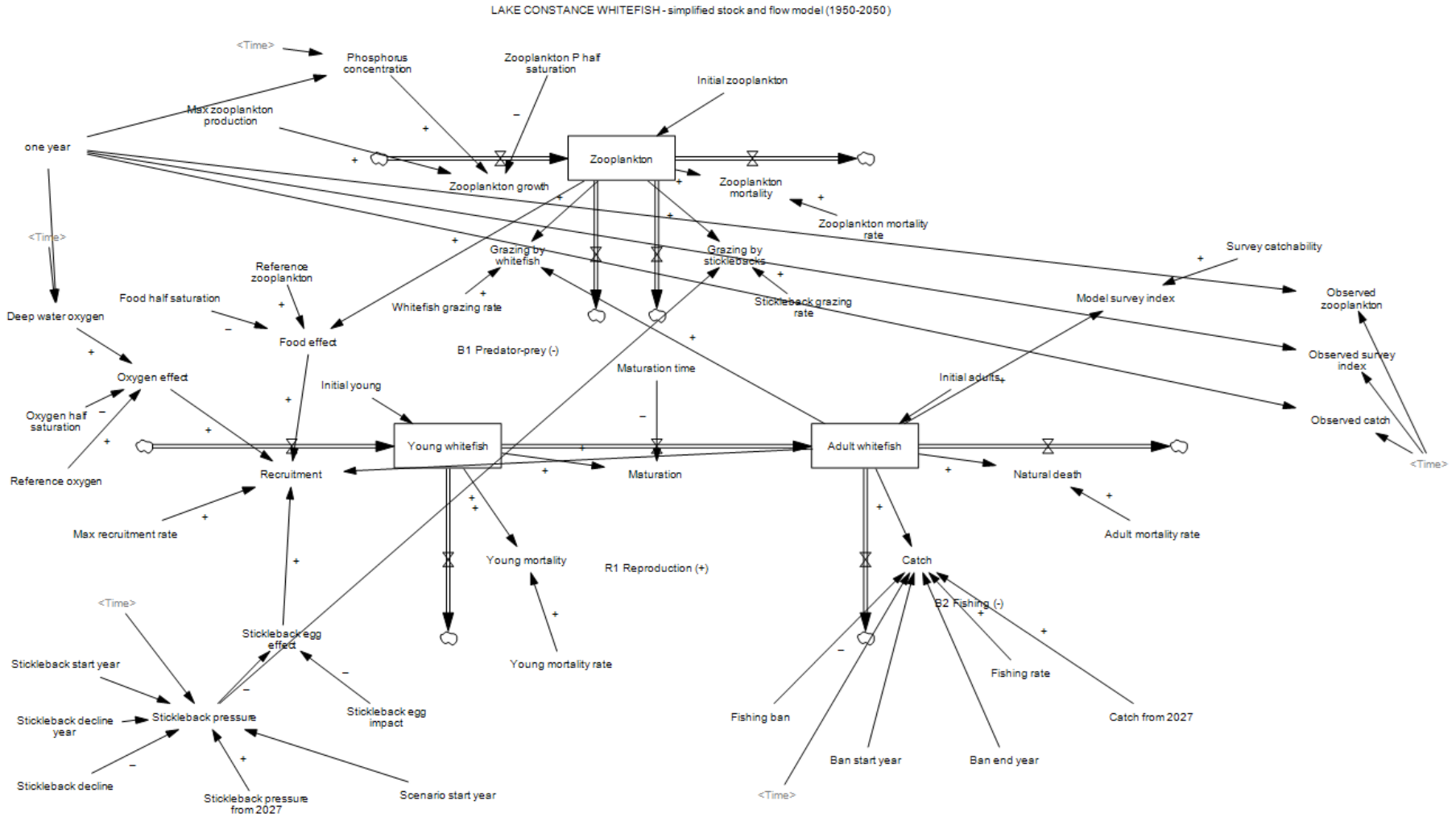
Hypothesis

- **H2**: The model can show the contribution effects of the 2024 fishing ban and the decline of the stickleback population to the recovery of the whitefish stock.

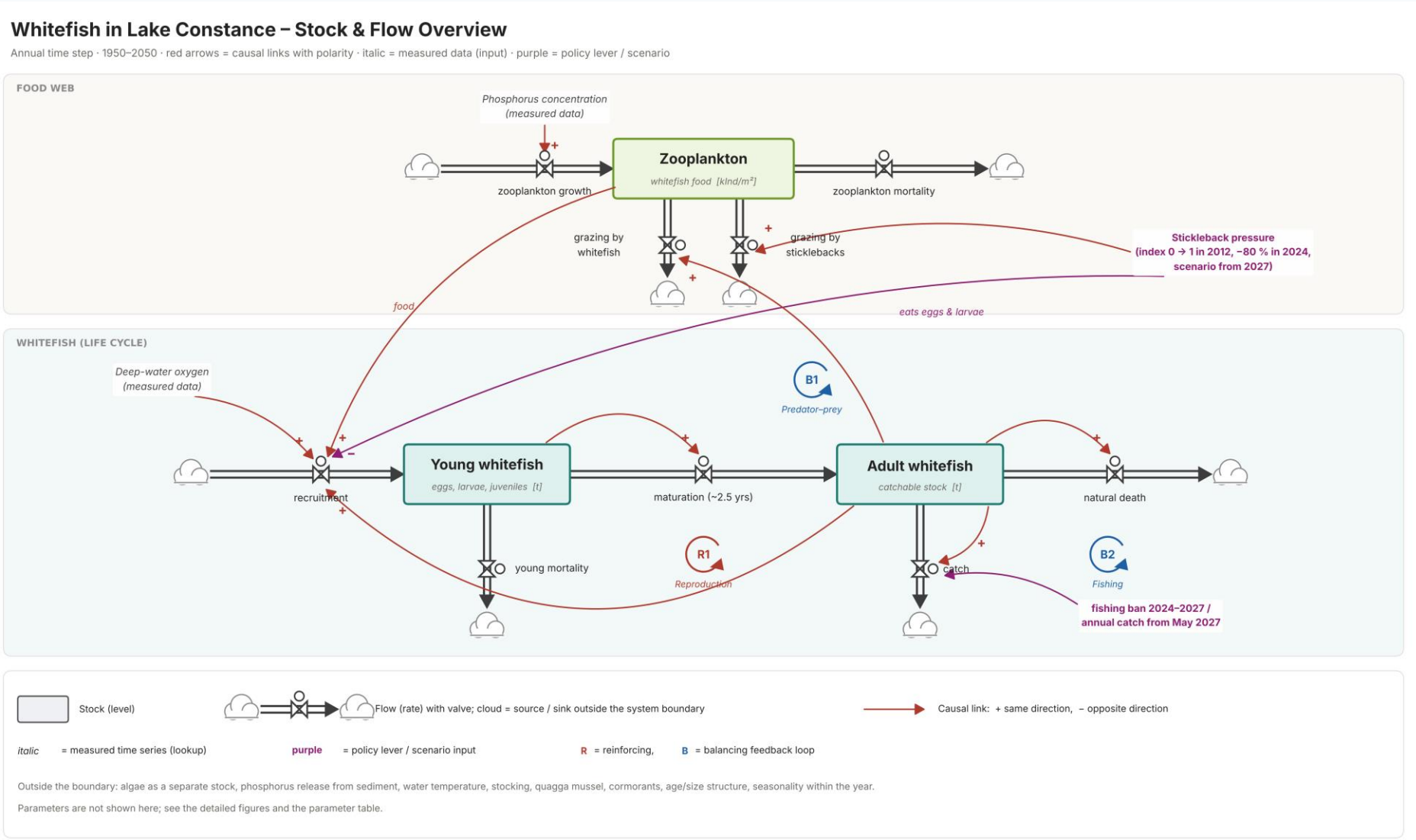
Hypothesis

- **H3**: The model can simulate the maximum annual catch that can be permitted from 2027 onwards, to ensure that the stock remains above a reference level until 2050
  - on low and renewed stickleback pressure

Model - Overview



Model - Overview



Model - Variables

State Variable	*Input Flows*	*Output Flows*
Adult Whitefish	Maturation	•Natural death •Catch
Young Whitefish	Recruitment	•Maturation •Young mortality
Zooplankton	Zooplankton growth	•Zooplankton mortality •Grazing by whitefish •Grazing by stickleback

Auxiliary Variables: Food effect, Oxygen effect, Stickleback egg effect, Stickleback pressure, Phosphorus concentration, Deep water oxygen, Observed catch

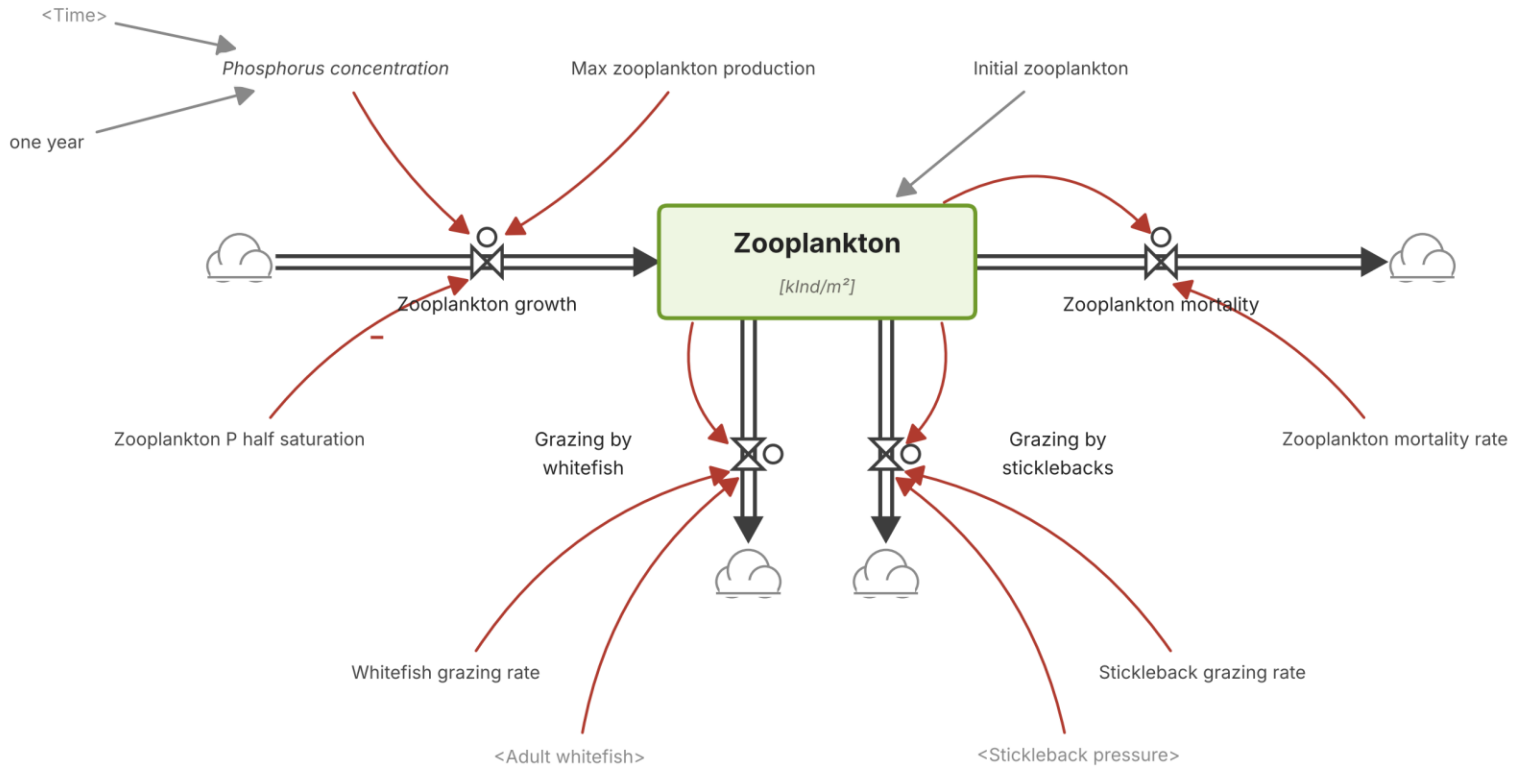
Model - ODE's

Variable	*ODE*
Adult Whitefish	$\frac{d\ Adult\ Whitefish}{dt} = Maturation - Natural\ Death - Catch$
Young Whitefish	$\frac{d\ Young\ Whitefish}{dt} = Recruitment - Maturation - Young\ Mortality$
Zooplankton	$\frac{d\ Zooplankton}{dt} = Zooplankton\ Growth - Zooplankton\ Mortality - Grazing\ by\ Whitefish - Grazing\ by\ Sticklebacks$

Model - Overview

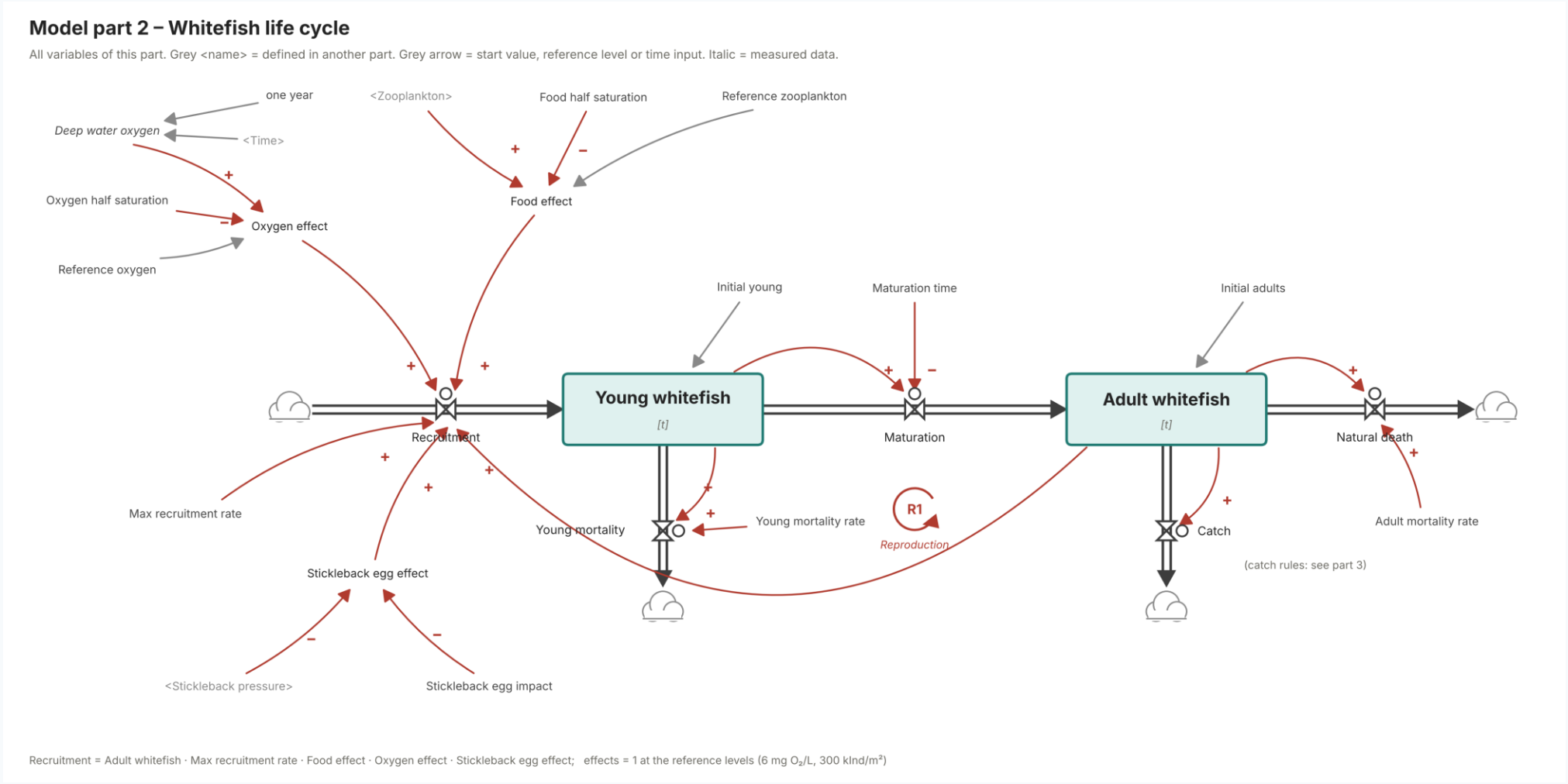
Model part 1 – Zooplankton (food web)

All variables of this part. All causal links are positive (+) unless marked –. Grey <name> = defined in another part. Italic = measured data.



Zooplankton growth = Max zooplankton production · P / (P + Zooplankton P half saturation) Grazing = Zooplankton · grazing rate · Adult whitefish (or Stickleback pressure)

Model - Overview

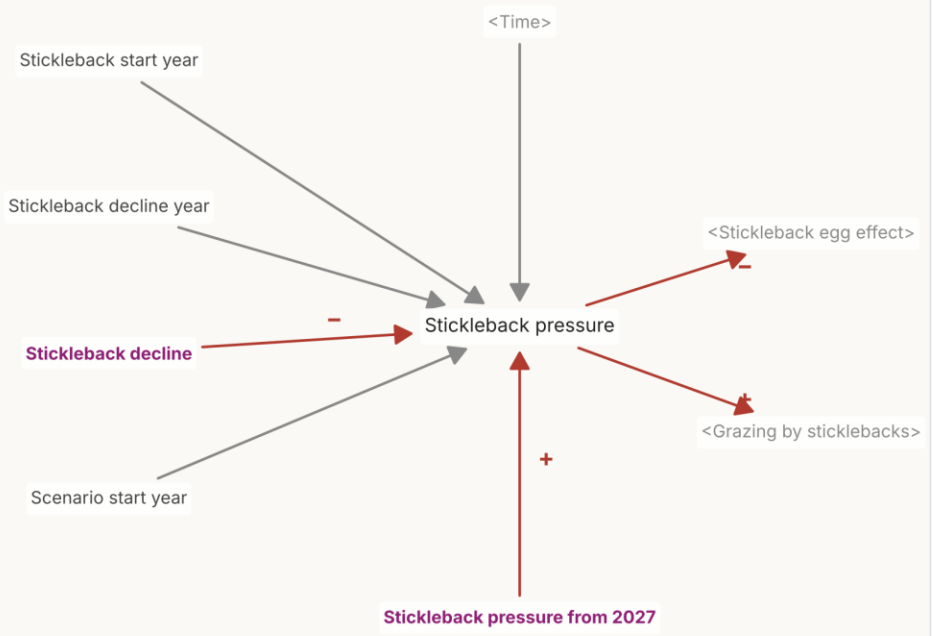


Model - Overview

Model part 3 – Stickleback pressure, fishing and outputs

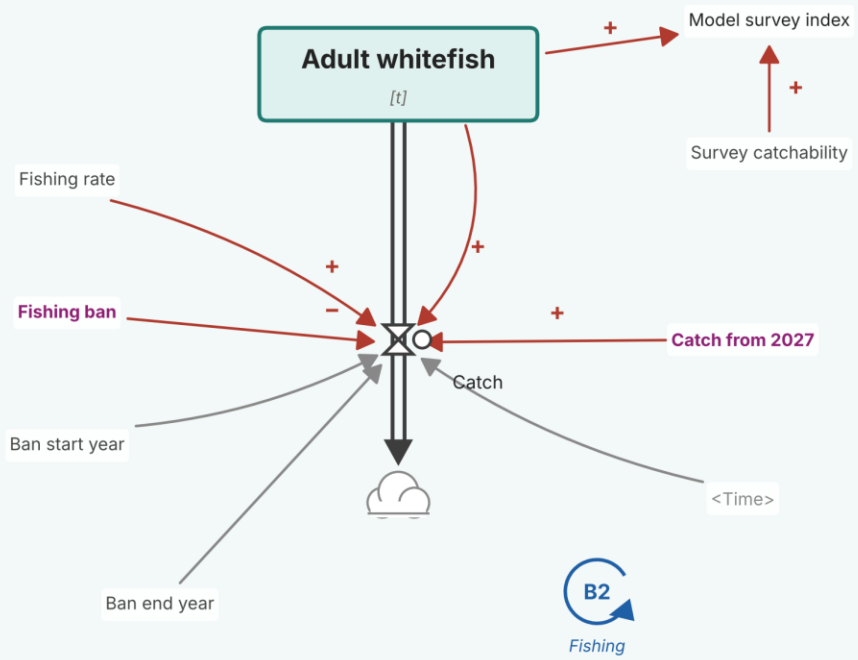
All variables of this part. Grey <name> = defined in another part. Grey arrow = time switch (no polarity). Purple = scenario lever.

STICKLEBACK PRESSURE (EXOGENOUS INDEX)



0 before 2012 · 1 from 2012 · 1 − 0.8 = 0.2 from 2024 · scenario value from 2027 (0.2 or 1)

FISHING AND OUTPUTS

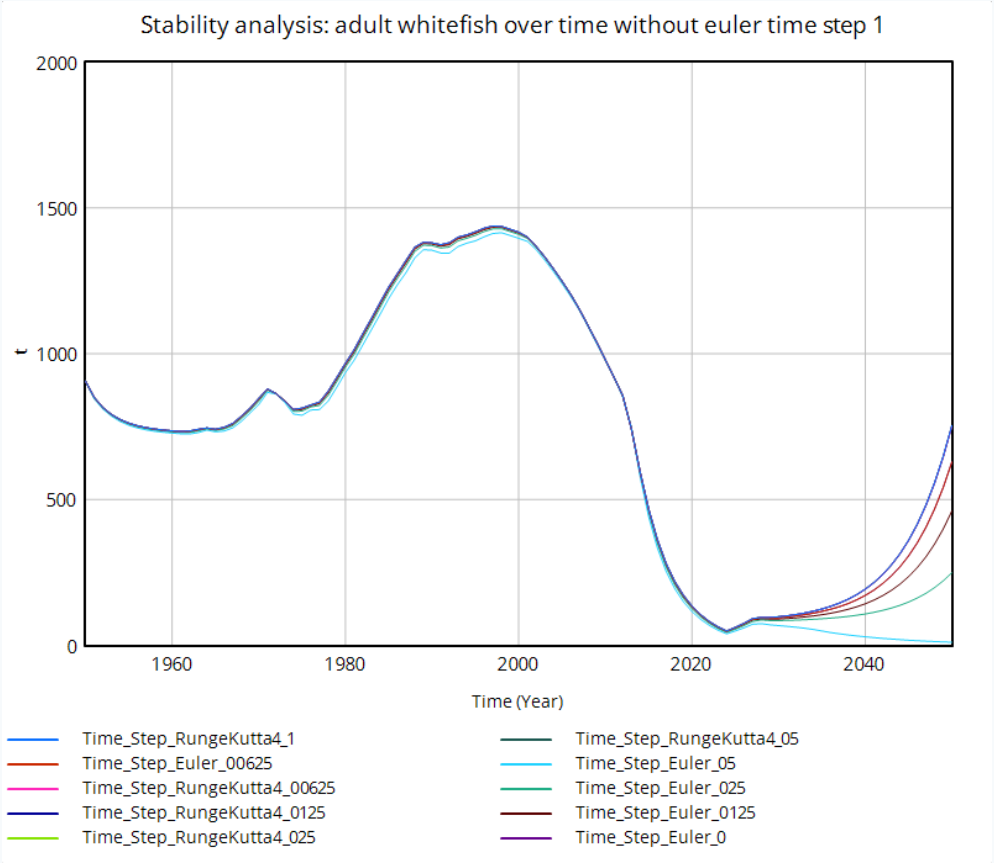


Catch: before 2024 = Fishing rate · Adult whitefish; 2024 – Apr 2027 = 0 (if Fishing ban = 1); from May 2027 = min(Catch from 2027, Fishing rate · Adult whitefish)

Model - Analysis

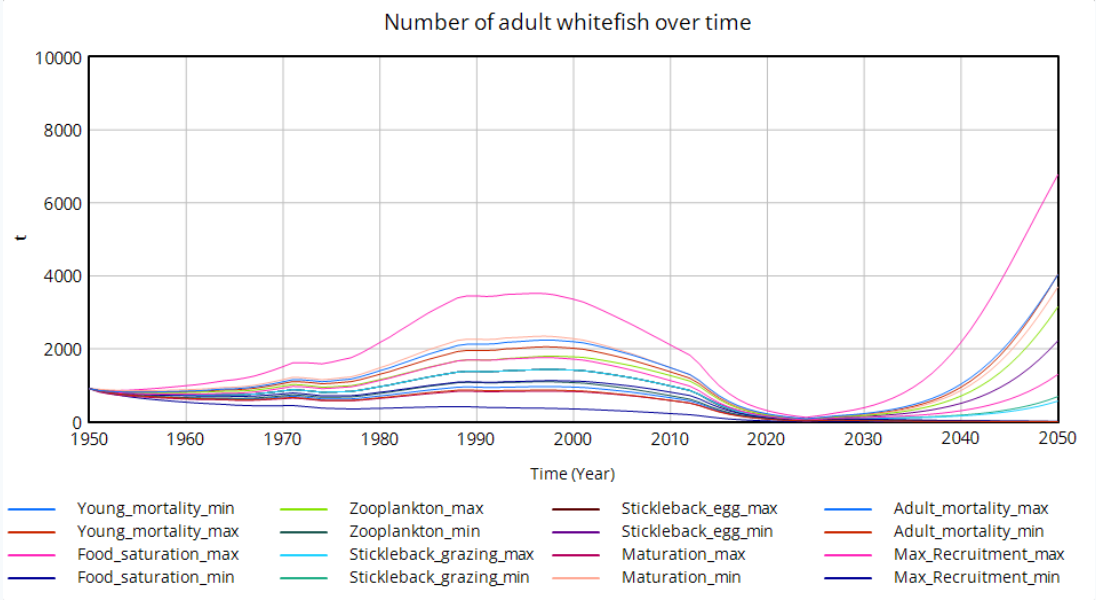
- Numerical stability
  - Euler and Runge-Kutta 4, different time steps (0.0625 to 1)
  - Historical phase is robust, projection until 2050 depends on the time step

- Validity
  - Units checked with Vensim
  - No decoupled variables
  - Assumptions and calibration separately



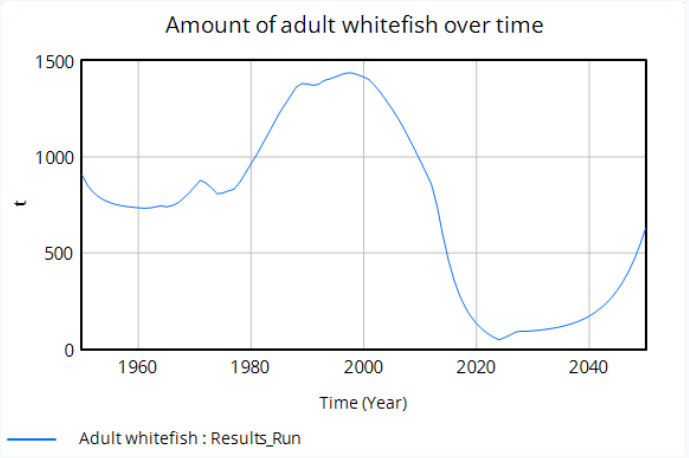
Model - Analysis

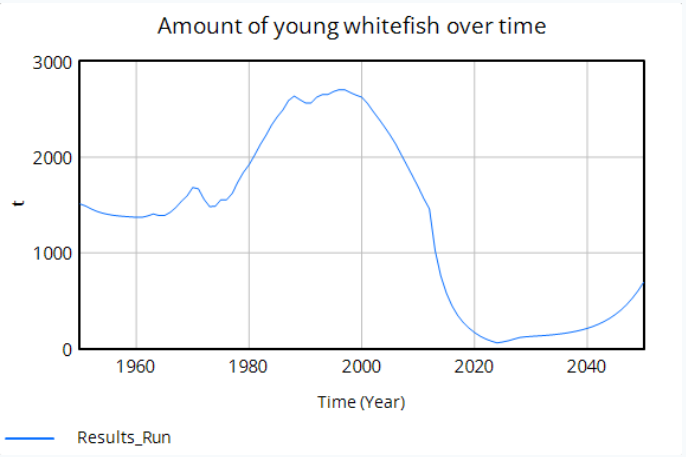
- Sensitivity
  - 8 parameters, ±10% each, without recalibration
  - Qualitative behavior is stable, but scale is sensitive

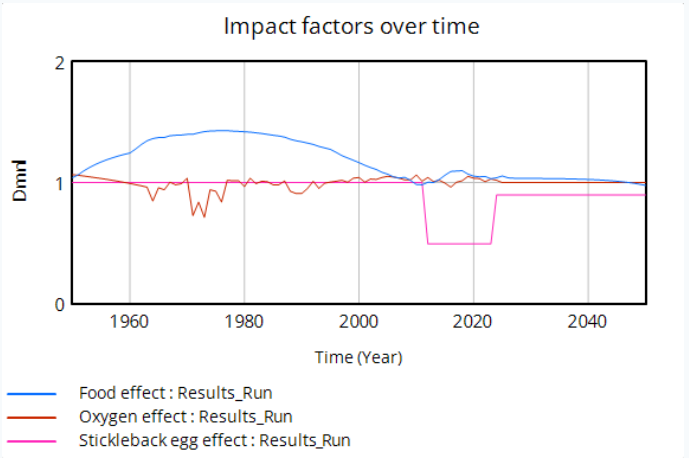


Results

- Growth through 1997, decline beginning in 2000, recovery only in the model
- Adults and young fish follow parallel trends
- Explanation in the model:
  - Food supply accounts for the increase and decline through 2012
  - Sticklebacks are an additional factor



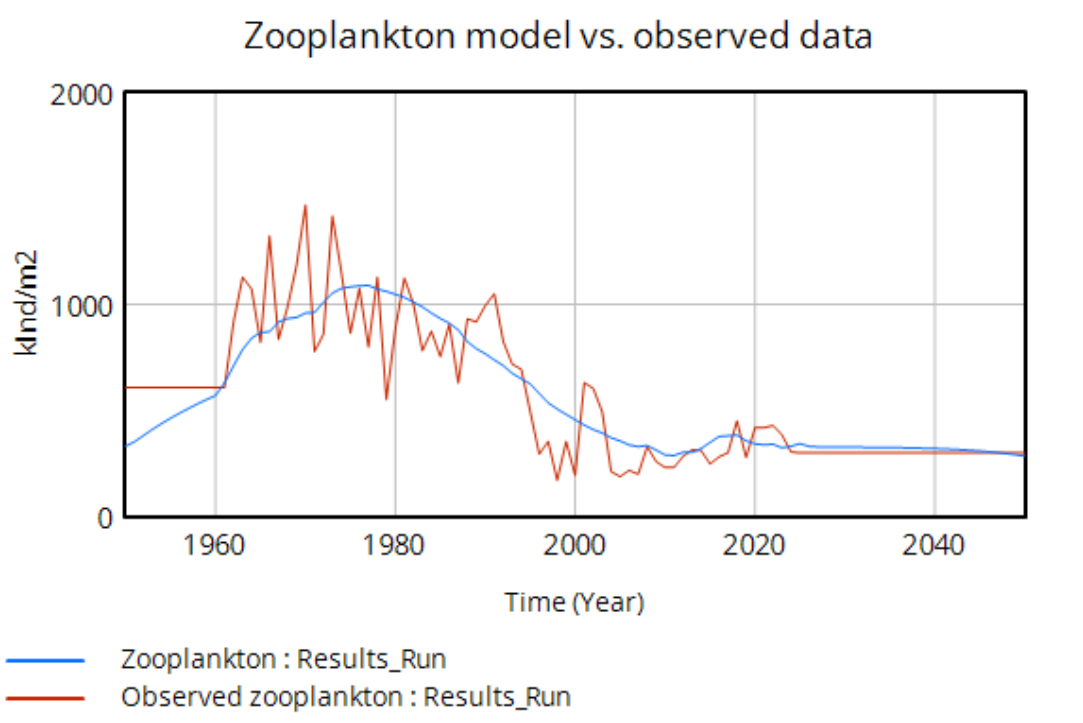


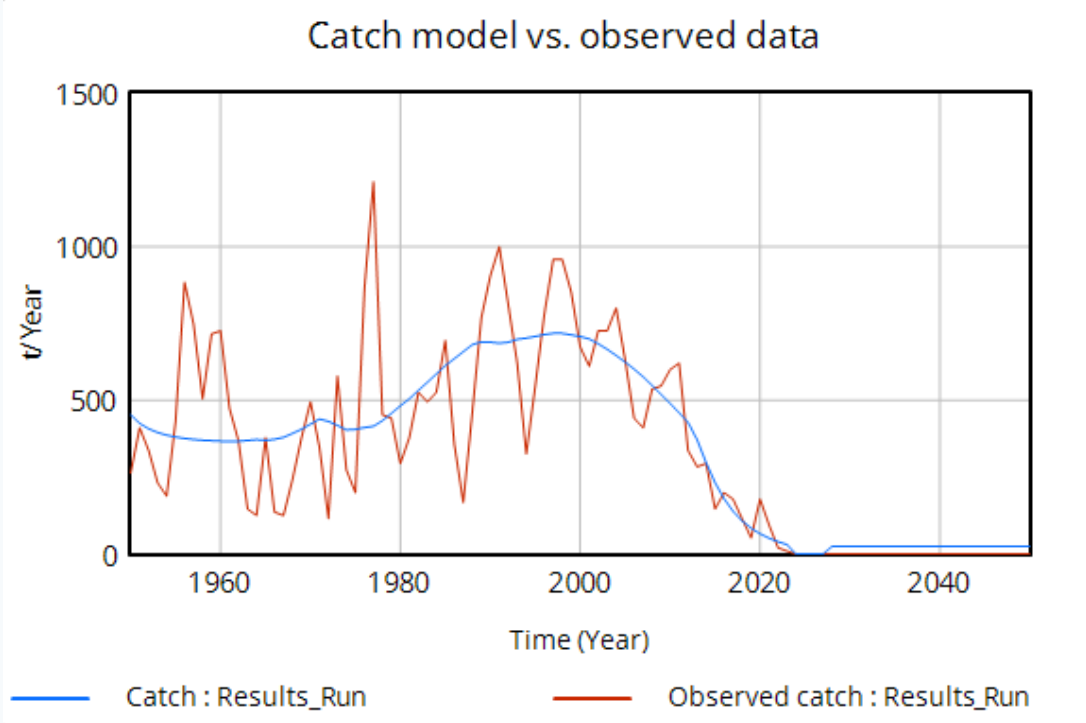


Results

H1: Reproducing the historical data

- Long-term trend and timing are right, but the peaks and annual fluctuations are not

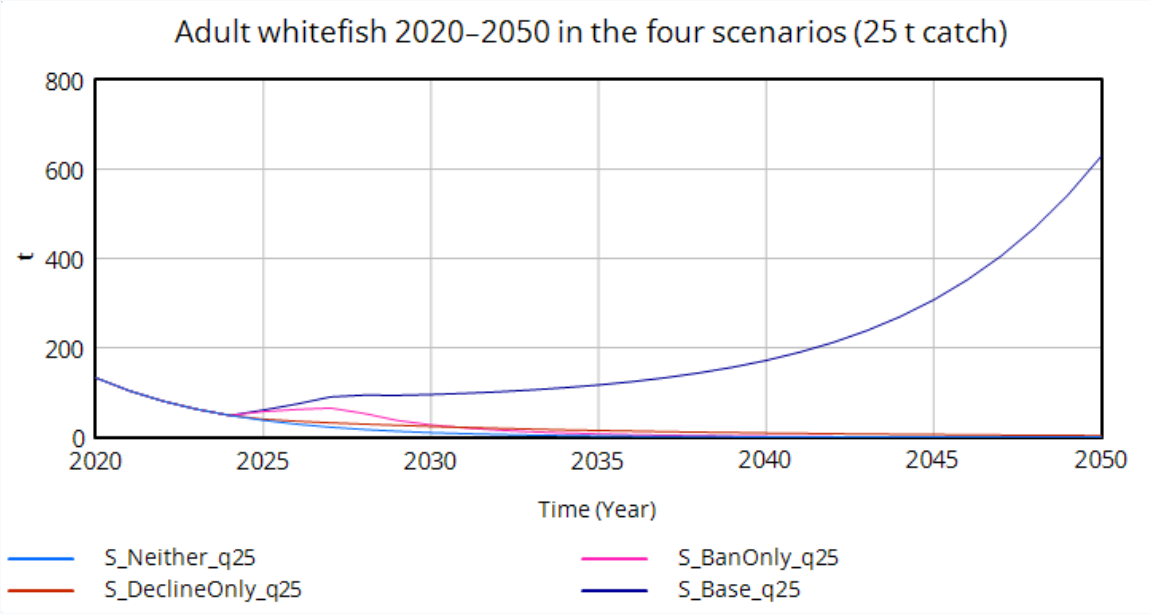


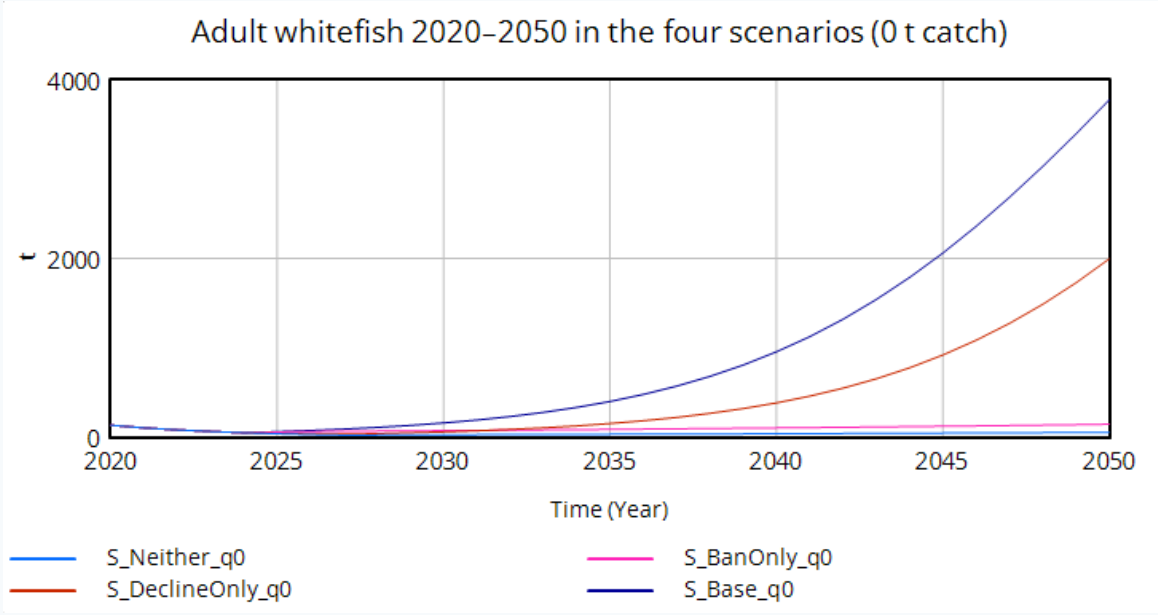


Results

H2: Fishing ban and decline of stickleback population contribution to the whitefish stock recovery

- 4 scenarios were plotted:
  - Ban and decline (baseline), ban only, decline only, and neither
  - Each with 0 t catch and 25 t catch per year
- Neither a fishing ban nor a decrease in the stickleback population is sufficient on its own; they have a synergistic effect

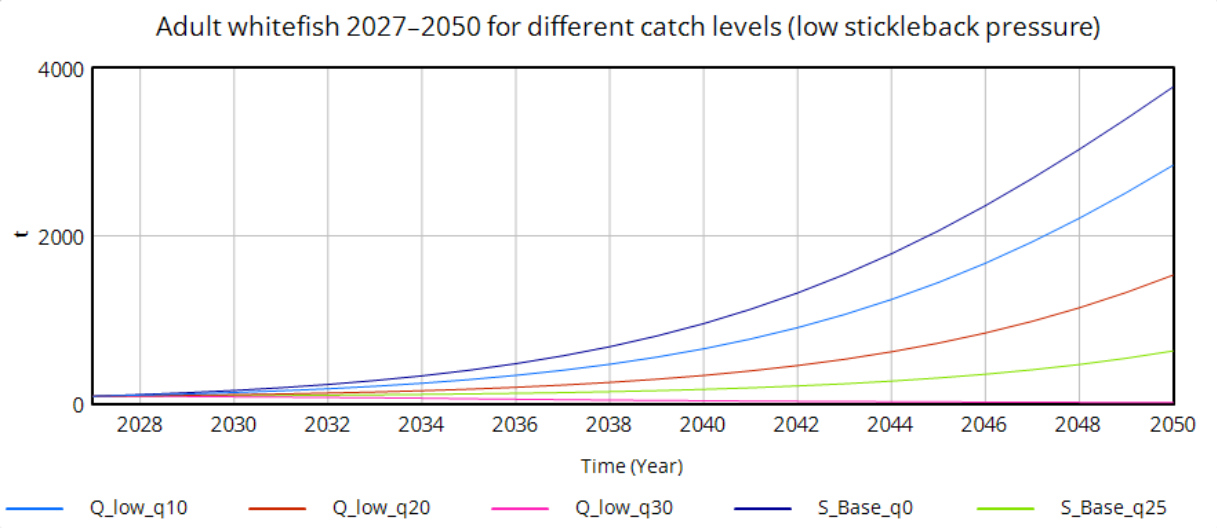


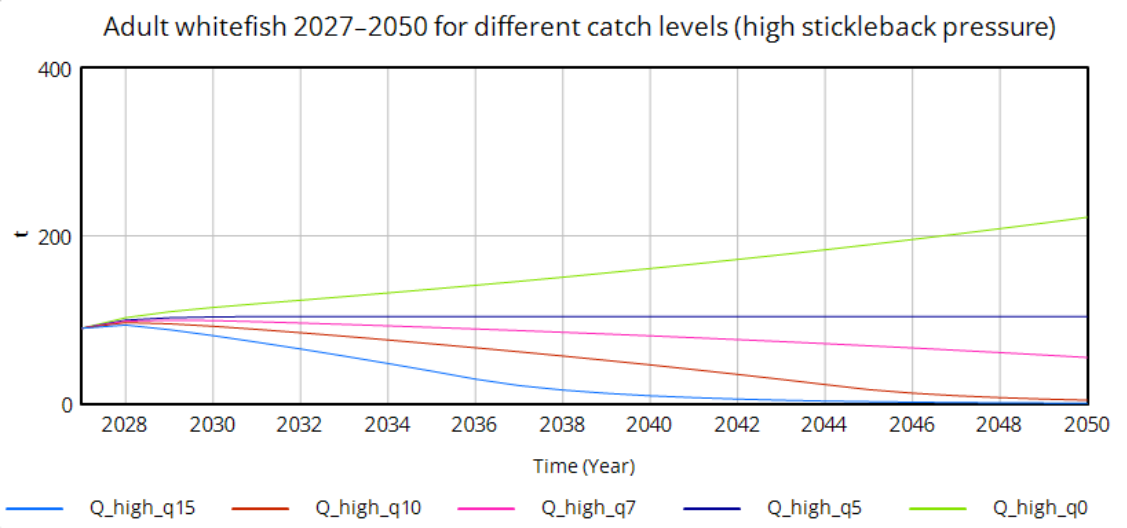


Results

H3: Calculation of the maximum annual catch from 2027 onwards to keep the stock above a reference level (50t) until 2050

- Different catch limits:
  - Approx. 27 t per year when stickleback pressure is low
  - Only approx. 5 t per year if stickleback pressure increases again





Open issues

- Water temperature not included
  - Scenarios through 2050 assume constant conditions → climate change
  - Warming could slow or accelerate recovery, the direction is uncertain
- Other factors not included
  - e.g. Quogga mussles, as a potential competition for food
- Data mainly for upper part of the lake

→ The model is a tool for comparing scenarios based on known drivers, but it is not a complete representation of the ecosystem

Q&A



Git Repository: https://github.com/the-nerd-sloth/co1-semester-project